

Sidhant Gurung

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EDUCATION

The University of Texas at Arlington

Expected Graduation: May 2027

Bachelor of Science in Electrical Engineering

- GPA: 3.68; Dean's List
- Relevant Coursework: Embedded Systems, Circuit Analysis II, Power System Stability & Control, Analog & Passive Signals

WORK EXPERIENCE

Qorvo, Applications Engineering Intern; Dallas, TX

May 2026- Present

- Designed an **RC miller cap** compensation network for a GaN **RF PA drain-switch driver**, reducing slew switching rates from **30ns** to **~5μs** to control inrush current and improve power-stage startup behavior
- Designing a 4-layer drone control PCB in **Altium Designer** integrating motor-controllers ICs and **CAN** communication to drive and coordinate four **BLDC motors**.
- Developed embedded firmware on **ARM Cortex-M0, M0+, M4** platforms, implementing **I2C, SPI, ADC**, and **UART** interfaces for peripheral control, sensor integration, and system monitoring.

Dr. Chenyun Pan's Research Group, Research Volunteer; Arlington, TX

April 2026 – Present

- Conduct literature reviews on spiking neural networks (SNNs), biologically plausible learning rules, and neuromorphic computing applications to support reinforcement learning research.
- Studying rewards-modulated spike-timing dependent plasticity (R-STDP) algorithms toward improving autonomous control performance on reinforcement learning benchmarks.

Pulsed Power and Energy Laboratory, Undergraduate Research Assistant; Arlington, TX

February 2026 – May 2026

- Tested and discharged 64V battery systems using a **bidirectional power supply**, monitoring current and voltage with multimeter to support energy storage characterization research.
- Fabricated **copper busbars** for battery interconnects by milling and sanding to specification, improving fitment and reducing low-resistance connection issues.
- Assembled and wired programmable power supplies to support pulsed power research setups.

PROJECTS

Haptic Focus Assistant

ESP32-C3, Bluetooth Low Energy (BLE), PWM, GPIO, Li-Po Power Systems, C++, KiCad

- Led hardware design on a 3-person team for an ESP32-based wearable system integrating **BLE** communication, PWM haptic control, Li-Po power, and audio feedback.
- Designed the schematic and 2-layer PCB in **KiCad**, integrating a **PWM**-driven haptic motor, piezo buzzer, and Li-Po power architecture with **flyback protection** and **signal integrity** considerations.
- Assembled, soldered and validated hardware using an **oscilloscope** and multimeter to verify motor operation and circuit continuity.

AI Heart Rate Tracker

ESP32 PPG Sensor Integration, GSR Sensing, I2C, ADC, Data Logging, Python

- Designing a biometric system integrating **PPG** and **GSR** sensors with an **ESP32** for stress and anxiety detection.
- Developing a real-time signal processing pipeline with digital filtering and peak detection targeting **95%+** heart rate accuracy at **10Hz** sampling.
- Using serial monitor output to debug sensor readings and verify **I2C** communication between the ESP32 and peripherals.

Temperature Dependent Speed Controller

Arduino IDE, C++, LTspice, Fusion 360, I2C, UART, SPI, PWM, AutoCAD

- Developed an Arduino-based temperature-controlled DC motor system in **C++** that modulates **PWM** duty cycle proportionally based on real-time thermistor readings.
- Achieved **±2°C** thermal regulation accuracy through iterative duty cycle threshold tuning across 20+ test runs.
- Created enclosure and PCB designs in **Fusion 360** and **AutoCAD** optimizing power distribution, trace routing, and airflow.

LEADERSHIP & COMMUNITY INVOLVMENT

IEEE Eta Kappa Nu (HKN) – Student Member

March 2026 – Present

- Inducted as a member based on top 25% academic standing, along with demonstrated leadership and character.

Additional Involvement: IEEE, IEEE-PES (Power and Energy Society), ACM, KCA

SKILLS

Languages: C, C++, MATLAB, Python.

Embedded Systems: GPIO, ADC, PWM, I2C, SPI, UART, BLE.

Hardware: Oscilloscope, Signal Conditioning, Sensors (PPG, GSR).

Design & Tools: Circuit Design & Analysis, PCB Layout, Control Systems, LTspice, KiCad, Lab Instrumentation, Soldering